

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

A. Shortest Path

time limit per test: 1 second
memory limit per test: 1024 megabytes

You are given an **directed weighted** graph with N nodes and M edges. The nodes are numbered from 1 to N . The graph contains no self-loops or multiple edges.

There is a source and a destination. Your task is to find the shortest distance from the source node to the destination node and print the path taken. If multiple shortest paths exist, print any one of them. If no such path exists, print -1 .

Input

The first line contains four integers $N, M, S, D (2 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, D \leq N)$ — the number of vertices, total number of edges, source and destination.

The second line contains M integers $u_1, u_2, u_3 \dots u_m (1 \leq u_i \leq N)$ — where the i -th integer represents the node that is one endpoint of the i -th edge.

The third line contains M integers $v_1, v_2, v_3 \dots v_m (1 \leq v_i \leq N)$ — where the i -th integer represents the node that is the other endpoint of the i -th edge.

The fourth line contains M integers $w_1, w_2, w_3 \dots w_m (1 \leq w_i \leq 10^6)$ — where the i -th integer represents the weight of the i -th edge.

The i -th edge of this graph is from the i -th node in the second line to the i -th node in the third line.

Output

- If a valid path exists from S to D , print the shortest distance S to D on the first line.
- On the second line, print the nodes in the path in order from S to D . If multiple shortest paths exist, print any one of them.
- If no such path exists, print -1 .

Examples

input	Copy
4 3 4 2 1 3 4 2 2 3 3 4 5	
output	Copy
9 4 3 2	

input	Copy
6 5 1 5 1 4 1 6 4 2 1 6 2 6 3 3 4 3 4	
output	Copy
-1	

CSE221 Assignment 07 Summer 2026

Finished

Contestant



→ Last submissions

Submission	Time	Verdict
388869133	Aug/30/2026 07:06	Accepted
388869101	Aug/30/2026 07:06	Accepted
388868703	Aug/30/2026 06:56	Accepted
388561456	Aug/27/2026 14:07	Accepted
388559707	Aug/27/2026 13:49	Time limit exceeded on test 68
388555500	Aug/27/2026 13:02	Wrong answer on test 8
388555263	Aug/27/2026 12:59	Wrong answer on test 2

input	Copy
<pre>2 1 2 1 2 1 7</pre>	
output	Copy
<pre>7 2 1</pre>	

input	Copy
<pre>5 7 2 4 1 1 5 4 2 3 2 5 4 3 5 5 4 3 3 8 2 6 6 4 3</pre>	
output	Copy
<pre>7 2 3 4</pre>	

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Sep/21/2026 18:39:06^{UTC+6} (i1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by



B. Where to Meet?

time limit per test: 2 seconds

memory limit per test: 1024 megabytes

Alice and Bob are in a hurry to meet each other and have to traverse through a **directed** graph with **weighted** edges. The nodes are numbered from 1 to N . The graph contains no self-loops or multiple edges.

Alice starts from node S and Bob starts from node T . They want to find a common node in the graph where they can meet each other in the minimum amount of time. Alice or Bob can wait at any node if they want to.

Input

The first line contains four integers N, M, S, T ($2 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, T \leq N$) — the number of vertices, the total number of edges, the starting node of Alice, and the starting node of Bob.

The next M lines will contain three integers u_i, v_i, w_i ($1 \leq u_i, v_i \leq N, 1 \leq w_i \leq 10^6$) — there is a edge from the node u_i to the node v_i with a weight w_i .

Output

Print two integers separated by a space: the minimum time required for Alice and Bob to meet, and the node where they meet. If there are multiple such nodes, print the smallest node. If no such node exists, print -1 .

Examples

input	Copy
<pre>5 5 1 5 1 2 1 2 3 1 5 3 2 1 4 2 5 4 2</pre>	
output	Copy
<pre>2 3</pre>	

input	Copy
<pre>6 5 1 5 1 2 3 4 1 3 1 6 4 6 2 3 4 6 4</pre>	
output	Copy
<pre>-1</pre>	

input	Copy
<pre>2 1 2 2 2 1 7</pre>	
output	Copy
<pre>0 2</pre>	

CSE221 Assignment 07 Summer 2026

Finished

Contestant

→ Last submissions

Submission	Time	Verdict
388636092	Aug/28/2026 07:33	Accepted
388572016	Aug/27/2026 15:55	Accepted
388571851	Aug/27/2026 15:53	Time limit exceeded on test 46
388571599	Aug/27/2026 15:51	Time limit exceeded on test 46
388570073	Aug/27/2026 15:35	Wrong answer on test 87
388569308	Aug/27/2026 15:28	Wrong answer on test 87

input

Copy

```
5 7 2 5
1 5 3
1 4 8
5 3 2
4 5 6
2 5 6
3 4 4
2 3 3
```

output

Copy

```
3 3
```

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Sep/21/2026 18:39:13^{UTC+6} (i1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by



C. Minimize the Danger

time limit per test: 2 seconds

memory limit per test: 1024 megabytes

You are in a city with N cities connected by M bi-directional roads. Each road has a danger level, where a higher number means the road is more dangerous.

You start in city 1 and need to go to every other city. The goal is to find the minimum danger level you would face to reach each city from city 1. The danger of a path is defined as the highest danger level of any road on that path.

For each city, find the minimum danger level of the path from city 1. If a city is not reachable from city 1, print -1 . The danger of reaching city 1 is always 0.

Input

The first line contains two integers $N, M (2 \leq N \leq 2 \cdot 10^5, 1 \leq M \leq 3 \cdot 10^5)$ — the number of cities, the total number of roads.

The next M lines will contain three integers $u_i, v_i, w_i (1 \leq u_i, v_i \leq N, 1 \leq w_i \leq 10^6)$ — node u_i is connected to node v_i with a weight w_i .

Output

Output n integers, where i -th integer represents the minimum danger level you'd have to face in order to go from city 1 to city i .

Examples

input	Copy
4 3 2 1 3 2 3 5 3 4 3	
output	Copy
0 3 5 5	

input	Copy
6 5 1 2 3 1 4 5 1 6 2 2 6 3 4 6 1	
output	Copy
0 3 -1 2 -1 2	

input	Copy
2 1 2 1 5	
output	Copy
0 5	

input	Copy
-------	------

CSE221 Assignment 07 Summer 2026

Finished

Contestant

→ Last submissions

Submission	Time	Verdict
388635556	Aug/28/2026 07:22	Accepted
388635188	Aug/28/2026 07:14	Wrong answer on test 2

```
5 7
1 5 3
1 4 2
5 3 1
5 4 6
5 2 5
3 4 4
3 2 8
```

output

Copy

```
0 5 3 2 3
```

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Sep/21/2026 18:39:23^{UTC+6} (i1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by



D. Beautiful Path

time limit per test: 2 seconds

memory limit per test: 1024 megabytes

You are given an **directed** graph with N nodes and M edges. The graph contains no self-loops or multiple edges. The edges have no weight. The nodes are numbered from 1 to N and have a weight. You need to find the cost of a path, if there is any, from node S to node D with the minimum cost. The cost of a path is the sum of the weights of the nodes in that path.

Input

The first line contains four integers N, M, S, D ($2 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, D \leq N$) — the number of vertices, total number of edges, source, and destination.

In the next line, there will be N integers $w_1, w_2, w_3 \dots w_n$ ($1 \leq w_i \leq 10^6$) separated by spaces — representing the weights of each node.

The next M lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — there is an edge from the node u_i to the node v_i .

Output

If a valid path exists from S to D , print the minimum cost of the path. Otherwise, print -1 .

Examples

input	Copy
4 3 1 2 3 4 5 4 1 2 3 2 4 3	
output	Copy
7	

input	Copy
6 5 5 3 3 3 4 3 4 1 1 2 4 1 1 6 6 2 4 6	
output	Copy
-1	

input	Copy
2 1 1 1 7 6 2 1	
output	Copy
7	

CSE221 Assignment 07 Summer 2026

Finished

Contestant



→ Last submissions

Submission	Time	Verdict
388661026	Aug/28/2026 12:42	Accepted

input

Copy

```
5 7 3 5
3 8 2 6 6
1 5
1 4
5 3
4 5
2 5
3 4
2 3
```

output

Copy

```
14
```

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Sep/21/2026 18:39:30^{UTC+6} (i1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by



E. Parity Edges

time limit per test: 1.5 seconds
memory limit per test: 1024 megabytes

You are given a **directed weighted** graph with N nodes and M edges. The nodes are numbered from 1 to N . The graph contains no self-loops or multiple edges.

Your task is to find the shortest distance from node 1 to node N , with an additional constraint: the path cannot contain two consecutive edges with the same parity (i.e., both even or both odd). If no such path exists, print -1 .

Input

The first line contains two integers $N, M (2 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5)$ — the number of vertices, total number of edges.

The second line contains M integers $u_1, u_2, u_3 \dots u_m (1 \leq u_i \leq N)$ — where the i -th integer represents the node that is one endpoint of the i -th edge.

The third line contains M integers $v_1, v_2, v_3 \dots v_m (1 \leq v_i \leq N)$ — where the i -th integer represents the node that is the other endpoint of the i -th edge.

The fourth line contains M integers $w_1, w_2, w_3 \dots w_m (1 \leq w_i \leq 10^6)$ — where the i -th integer represents the weight of the i -th edge.

The i -th edge of this graph is from the i -th node in the second line to the i -th node in the third line.

Output

Print a single integer — the minimum distance from node 1 to node N following the given constraint. If there is no valid path, print -1 .

Examples

input

Copy

4 3
1 3 2
4 4 3
3 4 5

output

Copy

3

input

Copy

6 5
1 4 1 6 4
2 1 6 2 6
3 3 4 3 4

output

Copy

4

input

Copy

2 1
2
1

CSE221 Assignment 07 Summer 2026

Finished

Contestant



→ Last submissions

Submission	Time	Verdict
388697390	Aug/28/2026 19:28	Accepted

7	
output	Copy
-1	
input	Copy
5 7	
1 1 4 5 2 3 2	
4 5 3 4 4 5 3	
3 8 2 6 6 4 3	
output	Copy
8	

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Sep/21/2026 18:39:36^{UTC+6} (i1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by



[HOME](#) [TOP](#) [CATALOG](#) [CONTESTS](#) [GYM](#) [PROBLEMSET](#) [GROUPS](#) [RATING](#) [EDU](#) [API](#) [CALENDAR](#) [HELP](#)
[PROBLEMS](#) [SUBMIT CODE](#) [MY SUBMISSIONS](#) [STATUS](#) [STANDINGS](#) [CUSTOM INVOCATION](#)

F. Shortest Path Revisited

time limit per test: 3 seconds

memory limit per test: 1024 megabytes

You are given a **bidirectional weighted** graph with N nodes and M edges. The nodes are numbered from 1 to N . The graph contains no self-loops or multiple edges.

There is a source and a destination. Your task is to find the cost of the second shortest path from the source node to the destination node. If no such path exists, print -1 .

Note: Second shortest path will be strictly greater than the shortest path

Input

The first line contains four integers

$N, M, S, D (2 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, D \leq N)$ — the number of vertices, total number of edges, source, and destination.

The next M lines will contain three integers $u_i, v_i, w_i (1 \leq u_i, v_i \leq N, 1 \leq w_i \leq 10^6)$ — there is an edge between the node u_i and the node v_i with a weight w_i .

Output

- If a valid path exists from S to D , print the cost of the second shortest path from S to D .
- If no such path exists, print -1 .

Examples

input	Copy
4 3 2 3 2 1 3 2 3 5 3 4 3	
output	Copy
11	

input	Copy
6 5 3 4 1 2 3 1 4 5 1 6 2 2 6 3 4 6 1	
output	Copy
-1	

input	Copy
2 1 2 2 2 1 5	
output	Copy
10	

input	Copy
--------------	----------------------

CSE221 Assignment 07 Summer 2026

Finished

Contestant



→ Last submissions

Submission	Time	Verdict
388734204	Aug/29/2026 08:09	Accepted

```
5 7 2 5
1 5 3
1 4 2
5 3 1
5 4 6
5 2 5
3 4 4
3 2 8
```

output**Copy****7**

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Sep/21/2026 18:39:46^{UTC+6} (i1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by



[HOME](#) [TOP](#) [CATALOG](#) [CONTESTS](#) [GYM](#) [PROBLEMSET](#) [GROUPS](#) [RATING](#) [EDU](#) [API](#) [CALENDAR](#) [HELP](#)
[PROBLEMS](#) [SUBMIT CODE](#) [MY SUBMISSIONS](#) [STATUS](#) [STANDINGS](#) [CUSTOM INVOCATION](#)

G. Classic Bellman Ford

time limit per test: 1 second

memory limit per test: 256 megabytes

You have a weighted directed graph G with N nodes ($1 \leq N \leq 10^3$) and M edges ($0 \leq M \leq 10^4$). The weights of the edges can be negative, but it is guaranteed that the graph does not contain self-loops, parallel edges, or negative cycles.

Find the weight of the shortest path from node 1 to node N . It is guaranteed that there is a path from node 1 to node N .

Input

The first line of the input contains T ($1 \leq T \leq 10^4$), the number of test cases. The input for each test case starts with two space-separated integers N and M , followed by M lines containing three space-separated integers u , v , and w ($1 \leq u, v \leq N$ and $-10^9 \leq w \leq 10^9$) denoting that there is a directed edge of weight w from u to v .

It is guaranteed that the sum of NM over all test cases is at most 10^7 .

Output

For each test case, output the weight of the shortest path.

Example

input

Copy

```
2
4 4
1 4 5
1 2 4
2 3 -5
3 4 2
2 1
1 2 5
```

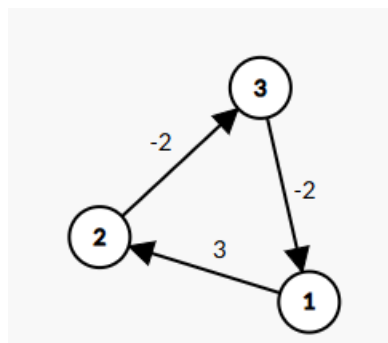
output

Copy

```
1
5
```

Note

A negative cycle is a cycle of negative total weight.



A negative cycle of weight $3 + (-2) + (-2) = -1$.

CSE221 Assignment 07 Summer 2026

Finished

Contestant



→ Last submissions

Submission	Time	Verdict
388751679	Aug/29/2026 12:15	Accepted
388742998	Aug/29/2026 10:18	Wrong answer on test 2

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Sep/21/2026 18:39:53^{UTC+6} (i1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by

